

BUDGET ESTIMATE

How the budget estimate works

What a dollar budget buys, derived from the real extraction pipeline — not a single model pass. Customer figures are US dollars (1 credit = \$1). Every figure is approximate and tunable.

01 The price we assume

Everything traces back to the cheapest model in our price book. Its per-million-token rates are the single underlying assumption.

Item	Value
Basis model	openai/gpt-oss-120b
Input price	\$0.09 / 1,000,000 tokens
Output price	\$0.36 / 1,000,000 tokens
Customer markup	x10 -> customer \$ (1 credit = \$1)

02 Filling the brain fans out

Each item runs the extraction pipeline. One agent finds the entities, then four agents run PER ENTITY — and every one re-reads the full item text. The body is sent to the model many times over.

Pipeline step	LLM calls	Reads
Identifier — find entities	1 / item	full item
Frontmatter — per entity	1 x N	full item
Judgment — relationship / status	1 x N	full item
Description — write / update	1 x N	full item + page
Timeline — per entity	1 x N	full item
Canonicalize / embed / graph	0 local	free

Total LLM calls per item = 1 + 4 x (entities found)

Because the body is resent every call, input tokens are roughly body x calls. The fan-out, not the body size, dominates cost.

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03 Per-item assumptions (tunable)

	Body tokens	Entities / item	Calls = 1+4N
1 email	2,000	4	17
1 document	12,000	8	33

04 Worked cost per item

$\text{input} = \text{body} \times \text{calls}$ $\text{output} = 180 \times \text{calls}$
 $\text{our cost} = \text{input}/1\text{e}6 \times 0.09 + \text{output}/1\text{e}6 \times 0.36$ $\text{price} = \text{cost} \times 10$

	Input tok	Output tok	Our cost	Price
1 email	34,000	3,060	\$0.00416	\$0.0416
1 document	396,000	5,940	\$0.03778	\$0.3778

05 Turning a budget into capacity

capacity = budget / price-per-item, both in dollars. The x10 markup is in numerator and denominator, so it cancels.

\$100 of remaining budget

~ **2,402 emails**

~ **264 documents**

Why this is far lower than a naive estimate

A single-pass guess gave ~18,500 emails per \$100. Modelling the real 17 calls/email — each re-reading the body — gives ~2,400: roughly 8x the cost for email, ~12x for documents.

Caveats

- The big lever is entities-per-item (4 / email, 8 / doc) and body size — estimates, not yet measured against the pilot's real corpus.
- Output is approximated at 180 tokens/call; input fan-out dominates, so it barely moves the total.
- Basis is the cheapest model; pricier routing is absorbed by the x10 markup.
- Embeddings (local fastembed) and canonicalization (deterministic) are free.
- Live billing meters every real call, so actual spend tracks this as the corpus is ingested.

Source: src/billing/metering.py (_pipeline_tokens, customer_cost_per_item)